

Encryption

At the end of this episode, I will be able to:

1. Encryption

Leamer Objective: *Learn about the latest encryption methods for wireless networks*

Description: In this episode, you will learn about the latest encryption methods used in wireless networks today. These are Wi-Fi Protected Access 2 (WPA2) and Wi-Fi Protected Access 3 (WPA3).

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- Wi-Fi Protected Access 2 (WPA2) is a security protocol and standard designed to secure wireless computer networks. It is the successor to the original WPA standard and was developed to address vulnerabilities found in its predecessor, particularly those associated with the use of the Temporal Key Integrity Protocol (TKIP).
 - WPA2 provides a higher level of security through the implementation of the Advanced Encryption Standard (AES) encryption algorithm.
 - WPA3 (Wi-Fi Protected Access 3) is the latest security protocol designed to enhance the security of wireless networks, succeeding WPA2.
 - WPA3 was developed by the Wi-Fi Alliance, the organization responsible for certifying Wi-Fi products. It introduces several improvements over its predecessor to address emerging security threats and provide stronger protection for wireless communications.
 - Here are key features of the WPA3 protocol:
 - Introduces the use of the Galois/Counter Mode (GCMP) protocol as an alternative to the Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP) used in WPA2. GCMP provides additional cryptographic strength.
 - Introduction of Individualized Data Encryption. In WPA2, a single encryption key is used for all devices connected to a Wi-Fi network. WPA3, on the other hand, provides unique encryption for each device, enhancing the security of data transmissions.
 - WPA3 includes measures to strengthen protection against brute-force attacks. Specifically, it introduces a Simultaneous Authentication of Equals (SAE) handshake, also known as Dragonfly, which is designed to resist password-guessing attacks more effectively.
 - WPA3 introduces the concept of forward secrecy, which means that even if an attacker were to compromise the key used to encrypt past communications, they would not be able to decrypt those communications. This is achieved through the use of Diffie-Hellman key exchange during the initial handshake.
 - WPA3 aims to improve resistance against offline dictionary attacks by making it more challenging for attackers to precompute potential passwords. This is achieved through the use of a more robust key exchange mechanism in the SAE handshake.
 - WPA3 includes a feature called "Wi-Fi Easy Connect" that simplifies the process of adding devices to a Wi-Fi network. It allows users to add devices to the network securely using a simpler method, such as scanning a QR code, without the need to manually enter a password.
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Additional Resources:

Wireless Security

<https://www.techtarget.com/searchnetworking/feature/Wireless-encryption-basics-Understanding-WEP-WPA-and-WPA2>